

Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT): A Fruit-Fly Connectome Toy Model

Version 3 — what the model does, and corrections

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Abstract

We study a toy model of a swarm of neural clusters, motivated by the *Drosophila* connectome [3]. Each cluster is a point in the plane; one step applies the operator pipeline $G = U \circ F \circ K \circ C$ of TO/TOGT (compress, clip, fold with a neighbour, rescale), with coupling α to the next cluster in a ring. Version 2 stated that for $|\alpha| < \frac{1}{2}$ the swarm map is a contraction with Lipschitz constant $L(\alpha) = \frac{1}{2} + |\alpha|$ and a unique fixed point, and that a pitchfork bifurcation occurs at $|\alpha| = \frac{1}{2}$. Version 3 shows that these statements do not hold: G is discontinuous, so it has no Lipschitz constant; a single agent already has four fixed points at $\alpha = 0$; and the quantity plotted as the bifurcation signature is fixed at 1 by construction. We then measure what the model does. It is *multistable*: for weak coupling every run settles fast, and 40 random starts reach 40 different end states. Near the threshold the version 2 paper proposed, its behaviour changes: in our runs the first cycles appear at $\alpha = 0.42$, most runs cycle at $\alpha = 0.5$ – 0.65 , and from $\alpha = 0.7$ some runs neither settle nor repeat within 400 steps. The corrections are proved in Lean 4 (Mathlib v4.32.0) with no `sorry`; the measurements are reproducible with the deposited code.

Keywords: TO/TOGT; operator pipeline; fruit-fly connectome; multi-agent dynamics; multistability; coupled map lattice; Lean 4. **MSC:** 37C25, 37N25, 92B20.

1 Relation to Prior Work

This paper supplements *Principia Orthogona, Volume One* [2] and the broader biological-transitions paper [1]. The FlyWire connectome [3, 4] and the larval connectome [5] motivate the ring-of-clusters topology only; no connectome data is fitted, and the model is not a quantitative neuroscience claim. Coupled systems of identical units can show multistability and switching between rest and oscillation, as in the analysis of coupled Hopf oscillators in perceptual bistability [6]. Bistable switching has also been modelled in the hypothalamic–pituitary–adrenal (HPA) stress axis [7]. Version 2 cited an HPA paper (Bhaskaran *et al.*, *J. Theor. Biol.* 540, 111077, 2022) whose DOI is not registered and which we could not find; it is withdrawn here.

2 The Operator Pipeline

Definition 1. For an agent state $s = (x, y) \in \mathbb{R}^2$ and its neighbour’s state n , one step is $G_\alpha(s; n) = U(F_\alpha(K_b(C(s)), n))$ with

- C (compress): keep the larger coordinate and set the other to 0; $C(x,y) = (x,0)$ if $|y| \leq |x|$, else $(0,y)$;
- K_b (clip): $K_b(s) = \text{clip}(s + b \text{sgn}(s), -1, 1)$ coordinatewise, $b = 0.1$, with $\text{sgn}(0) = +1$ as in the code `(np.sign(s + 1e-12))`;
- F_α (fold): $F_\alpha(s,n) = (1 - \alpha)s + \alpha n$;
- U (unfold): $U(s) = s/\|s\|$ for $s \neq 0$.

The swarm is N agents on a ring, $s_i^{t+1} = G_\alpha(s_i^t; s_{i+1}^t)$ (indices mod N). Version 2 added Gaussian noise of size $\sigma = 0.01$; all measurements below are noise-free.

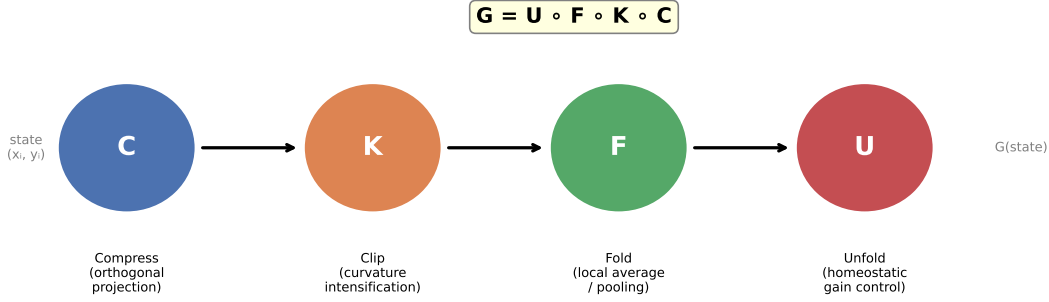


Figure 1: Schematic of $G = U \circ F \circ K \circ C$ (from version 2).

3 What Does Not Hold

Proposition 1 (No Lipschitz constant). *C sends $(1, 0.999)$ and $(0.999, 1)$, which are 0.002 apart in ℓ^1 , to $(1, 0)$ and $(0, 1)$, which are 2 apart. Hence C is discontinuous, and G_α has no Lipschitz constant for any α . At $\alpha = 0.3$ with neighbour $(0.6, 0.8)$ the two inputs are sent 1.32 apart.*

In version 2, $L(\alpha) = \frac{1}{2} + |\alpha|$ is a Lean definition; the theorems about it (for example $L(0.3) = 4/5$ and $(4/5)^6 < 27/100$) are arithmetic about that definition and are not connected to G .

Proposition 2 (Four fixed points). *Let $r = \sqrt{1.01}$. At $\alpha = 0$ the one-agent map $s \mapsto G_0(s; s)$ has the four distinct fixed points $(1, 0.1)/r$, $(-1, 0.1)/r$, $(0.1, 1)/r$, $(0.1, -1)/r$.*

Proof. For $(1, 0.1)/r$: C gives $(1/r, 0)$; K gives $(\text{clip}(1/r + 0.1), 0.1) = (1, 0.1)$ since $1/r \geq 0.9$; U returns $(1, 0.1)/r$. The other three are the same computation. $\square$

Version 2's formula gives $L(0) = \frac{1}{2}$, a strong contraction, which would force a unique fixed point.

Proposition 3 (The plotted signature is constant). *For $s \neq 0$, $\|U(s)\|^2 = 1$. So the mean of $\|s_i\|^2$ after any number of noise-free steps is exactly 1 for every α .*

Version 2's figure of this quantity against α therefore could not show a bifurcation; its 80 values lie in $[0.9989, 1.0010]$, a spread of 0.0021 produced by the added noise. Version 2's three “open obligations” contain no sorry: two conclude True, and the third holds because the swarm step is an identity placeholder.

α	different end states	period of each run (number of runs)
0–0.35	40	1 (all 40)
0.40	33	1 (40)
0.42	27	1 (36), 24 (4)
0.45	10	1 (36), 16 (4)
0.50	26	1 (17), 16 (21), 24 (2)
0.60	28	1 (13), 14 (16), 16 (11)
0.70	22	1 (22), 12 (4), none within 400 steps (14)
0.90	16	1 (28), 9 (1), none within 400 steps (11)
0.95	20	none within 400 steps (40)

Table 1: Attractors of the noise-free swarm, $N = 8$, 40 starts, 400 steps.

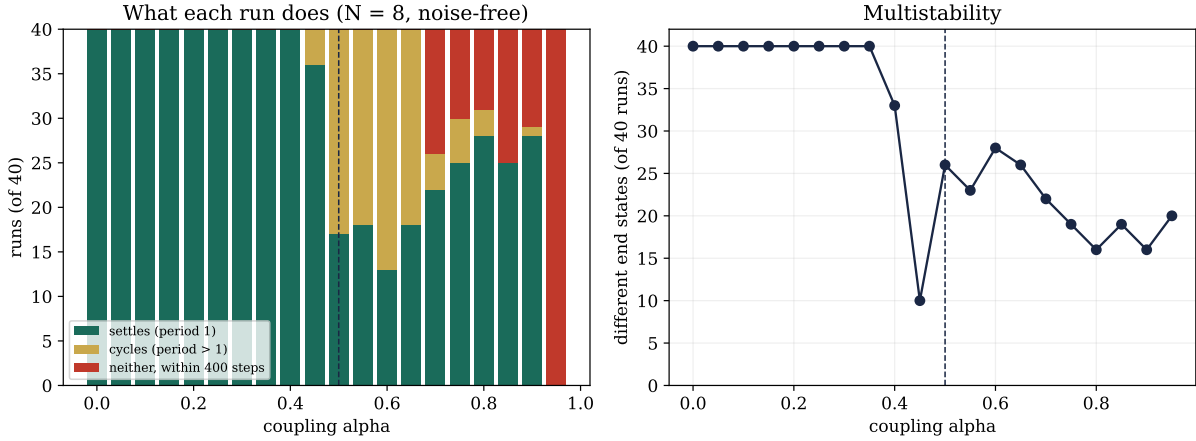


Figure 2: Left: what each of 40 runs does, by coupling. Right: number of different end states.

4 What the Model Does

For each α we ran $N = 8$ agents from 40 random starts (seeds 0–39, states uniform in $[-1, 1]^2$) for 400 steps, and recorded the number of different end states (rounded to 10^{-4}) and whether each run settles (period 1), cycles (period $p > 1$, detected up to $p = 39$) or does neither within the budget.

Three regimes appear. For $\alpha \leq 0.35$ the model is multistable: every run settles, and every run settles somewhere different. Settling is fast: at $\alpha = 0.3$, after six steps each of 200 runs is within 1.5% of its own end state (Figure 3), well inside version 2’s bound of 27%. From $\alpha = 0.42$ cycles appear, and for $0.5 \leq \alpha \leq 0.65$ more runs cycle than settle. From $\alpha = 0.7$ some runs neither settle nor repeat within 400 steps, and at $\alpha = 0.95$ none do. The change near $\alpha \approx 0.42$ – 0.5 is close to the threshold version 2 proposed, but it is a move from many resting states to cycles and irregular motion, not one state splitting in two. Why it happens there is not proved here.

Remark 1 (Interpretation). Multistability is the feature the multi-orbit picture needs: separate runs, or separate clusters, can end in separate stable states. The move to cycles as coupling grows is reminiscent of the switch between rest and oscillation in stress-axis models [7], but the model has not been compared with any physiological parameter, and the analogy is not a claim.

5 Lean 4 Verification

The file `BioSwarmCheck.lean` defines C , K_b , F_α , U and G exactly as in Definition 1 and proves, with no sorry and only the axioms `propext`, `Classical.choice`, `Quot.sound` (Lean 4.32.0, Mathlib v4.32.0; 17 declarations audited):

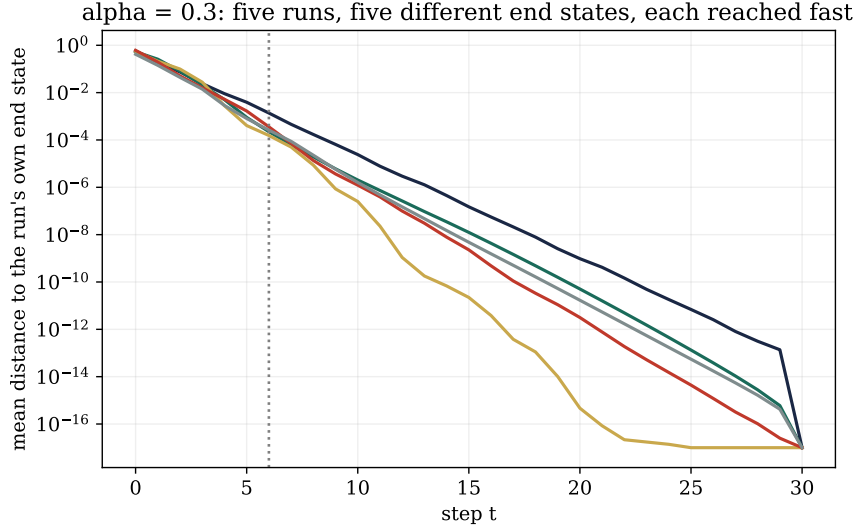


Figure 3: $\alpha = 0.3$: five runs, each approaching its own end state; the dotted line marks six steps.

Lean theorem	Statement
<code>C_jumps</code>	Proposition 1 (the ℓ^1 distances $1/500$ and 2)
<code>four_fixed_points</code>	Proposition 2
<code>U_norm</code>	Proposition 3

The measurements of Section 4 are numerical and are not proved; `multi_orbit_bioswarm_v3.py -check` reproduces them.

6 Open Questions

1. **The transition.** Why do cycles appear near $\alpha \approx 0.42$, and how does the onset depend on N and b ?
2. **Counting attractors.** How many stable states are there for weak coupling, as a function of N ? The runs suggest very many (every one of 40 starts ends differently).
3. **Connectome data.** Fit coupling constants to the FlyWire weight matrix [3] and compare with module structure [4].
4. **Stress axis.** Relate the transition to measurable HPA parameters [7], or drop the analogy.

7 Corrections since Version 2

(i) The Lipschitz constant $L(\alpha) = \frac{1}{2} + |\alpha|$, the contraction for $|\alpha| < \frac{1}{2}$ and the unique fixed point are withdrawn (Propositions 1, 2). (ii) The pitchfork figure is withdrawn and replaced by Figure 2 (Proposition 3). (iii) The six-step bound $(4/5)^6$ is replaced by the measured settling of Figure 3. (iv) The Lean file is replaced by `BioSwarmCheck.lean`; version 2's file stays in the record. (v) Reference [11] of version 2 is withdrawn and replaced by [7]. (vi) The dm^3 normalisation triple, which version 2 checked as arithmetic but did not use, is omitted.

Data and Code Availability

Deposited with this PDF: `multi_orbit_bioswarm_v3.tex`; `BioSwarmCheck.lean` and its axiom report; `multi_orbit_bioswarm.py` (version 2 operators, unchanged) and `multi_orbit_bioswarm_v3.py` (measurements, figures, -check); figures/ with `scan_v3.csv`.

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